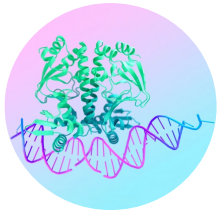
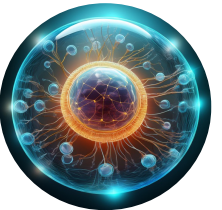


AI Agents for Biomedical Research – From Analysis to Clinical Decision Support



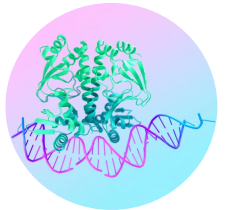
Mahbub Ul Alam
Data Engineer (AI)
SciLifeLab Data Centre
mahbub.ul.alam@scilifelab.se



By the end of this seminar, **you will...**



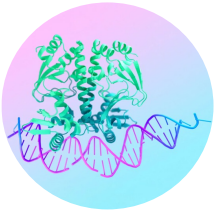
1. **Understand** what AI agents are and why they matter for research
2. **Know** which agent to choose for your specific workflow
3. **Be aware of** best practices for responsible use
4. **Have a glimpse** into the future of AI agents in biomedical research
5. (🎁 **Bonus!**) **Explore an interactive AI agent landscape tool** 😊



Outline



1. LLMs vs. AI agents: a critical distinction



2. Current AI agents in biomedical research workflows



3. How to get started with these tools



4. Emerging trends & what's next

From raw LLMs to AI agents: a comparison



Feature	Raw LLM	Chat interface	AI agent
What it is	Base model that predicts next tokens	LLM wrapped with UI, memory, and safety layers	LLM-powered system that autonomously pursues goals
Example	GPT-4 base	ChatGPT, Gemini	We'll see soon 😊
Data privacy & security	Data sent to external provider; limited control over sensitive information	Provider-controlled; data may be logged, used for training, or stored externally	Can run locally or on-premise; full control over data flow, tool permissions, and which systems access sensitive information

From raw LLMs to AI agents: a comparison



Feature	Raw LLM	Chat interface	AI agent
Interaction mode	Single input → single output	Multi-turn conversations with context retention	Goal-directed, iterative task execution
Memory	Stateless between calls	Session-based context window	Can maintain short and long-term memory across tasks
Tool access	None	Often includes search, file upload, code execution	Extensive, APIs, databases, browsers, external systems

From raw LLMs to AI agents: a comparison



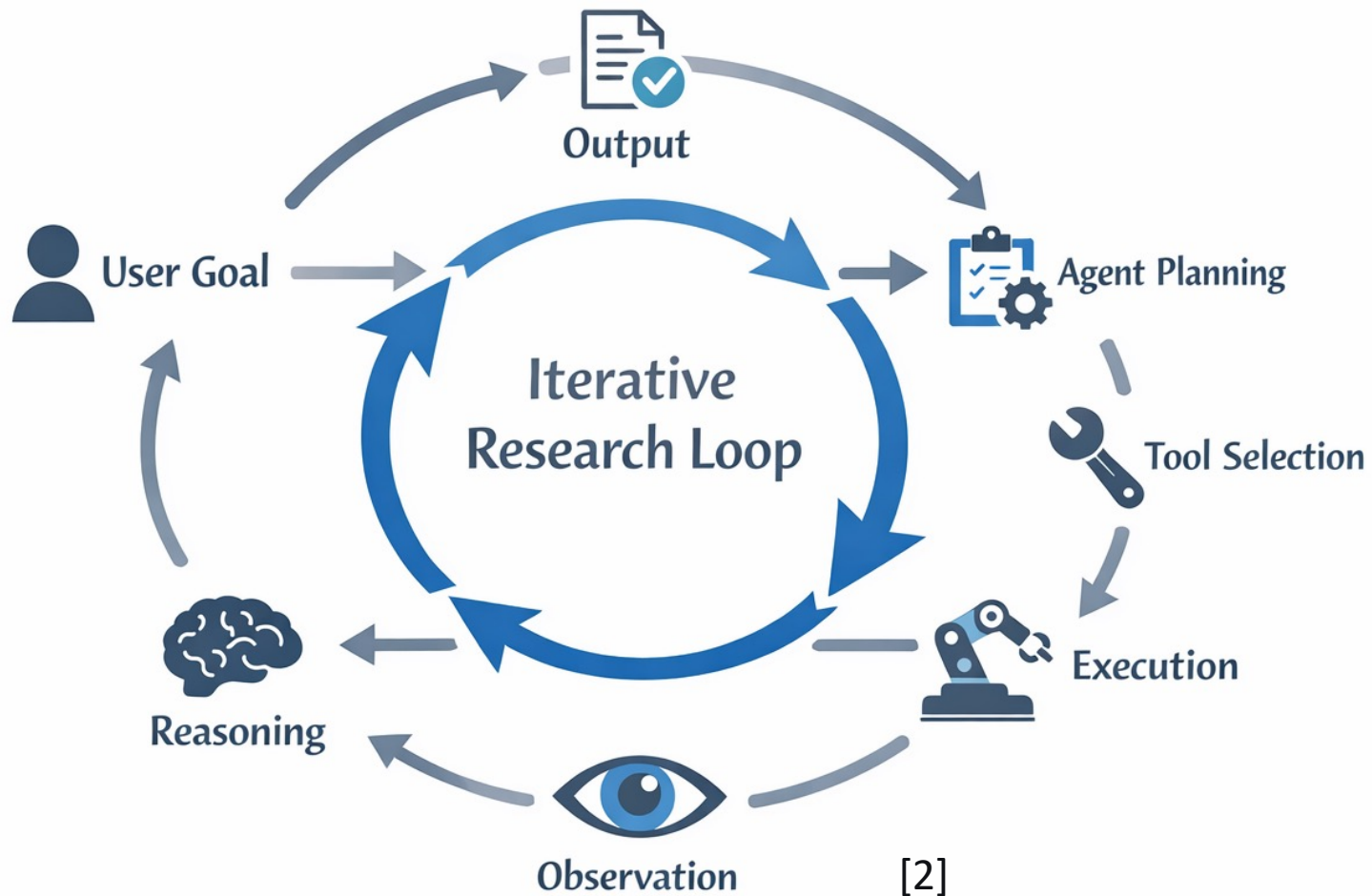
Feature	Raw LLM	Chat interface	AI agent
Decision making	None	Limited to response generation	Breaks down goals, selects tools, handles errors, iterates
Best for	Embedding in custom applications	Interactive Q&A, writing, analysis	Complex multi-step tasks requiring real-world actions
Autonomy	Zero, only responds when prompted	Reactive, waits for user input each turn	Proactive, plans, decides, and acts independently

Levels of autonomy in AI agents ^[1]



Level	Role	Key capabilities	Human involvement
0	No AI agents	ML models perform predefined tasks; no adaptation	Scientist does everything: hypothesis, tasks, execution
1	Assistants	Simple hypotheses; narrow protocol design; strong reasoning in selected tasks	Scientist defines hypothesis and tasks; AI executes
2	Collaborators	Hypotheses from data trends; rigorous protocols; interprets findings within broader context	Scientist proposes, AI refines; AI plans and executes tasks
3	Scientists	Creative, de novo hypotheses; develops new methods; continuous learning and adaptation	Minimal; AI drives discovery autonomously

AI agents in the scientific research loop



Autonomous iteration...
- from goal to discovery

Continuous refinement...
- output informs next planning cycle

Why not just using raw LLMs directly instead?



Raw LLM	AI agent (better alternative)
Risk of hallucinations	Action & verification
Creative brain, no grounding	Grounded research partner
No built-in fact checking	Executes & reports with sources
“ Garbage in, garbage out ”	Integrates live, validated data

“ An agent facilitates interfacing with LLMs in the biomedical space, while at the same time **safeguarding their functionalities** through sensible constraints! ” [3]

Hallucination in action: an example



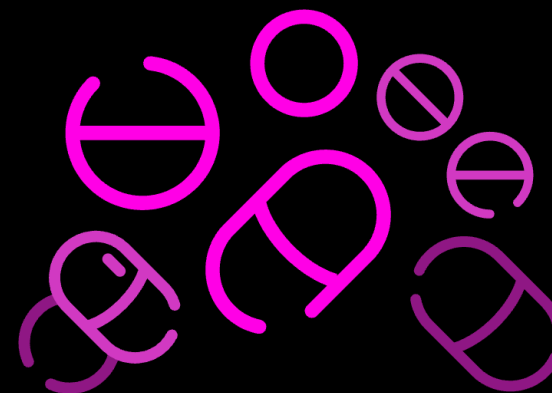
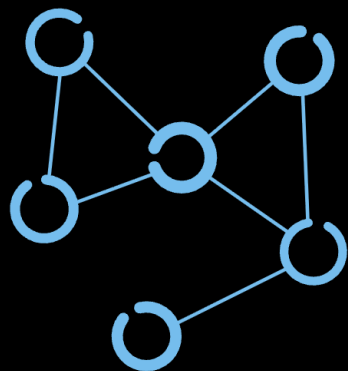
Raw LLM	AI agent (better alternative)
✗ “ The gene BRCA1 is located on chromosome 13 and is primarily involved in DNA repair ... ”	✓ Agent queries the NCBI Gene database → retrieves verified record “ BRCA1 is located on chromosome 17q21.31 (Source: <i>NCBI Gene, ID 672</i>) ”



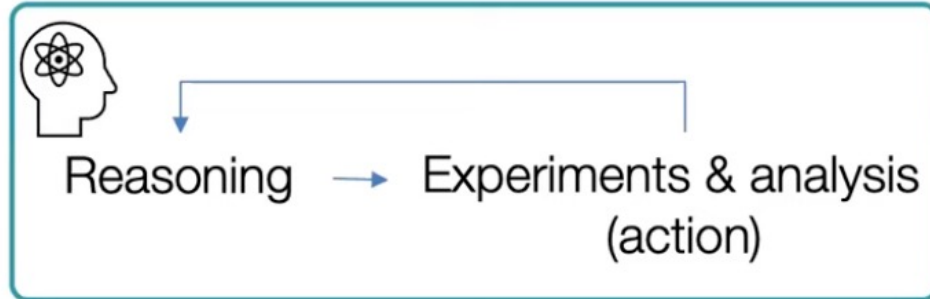
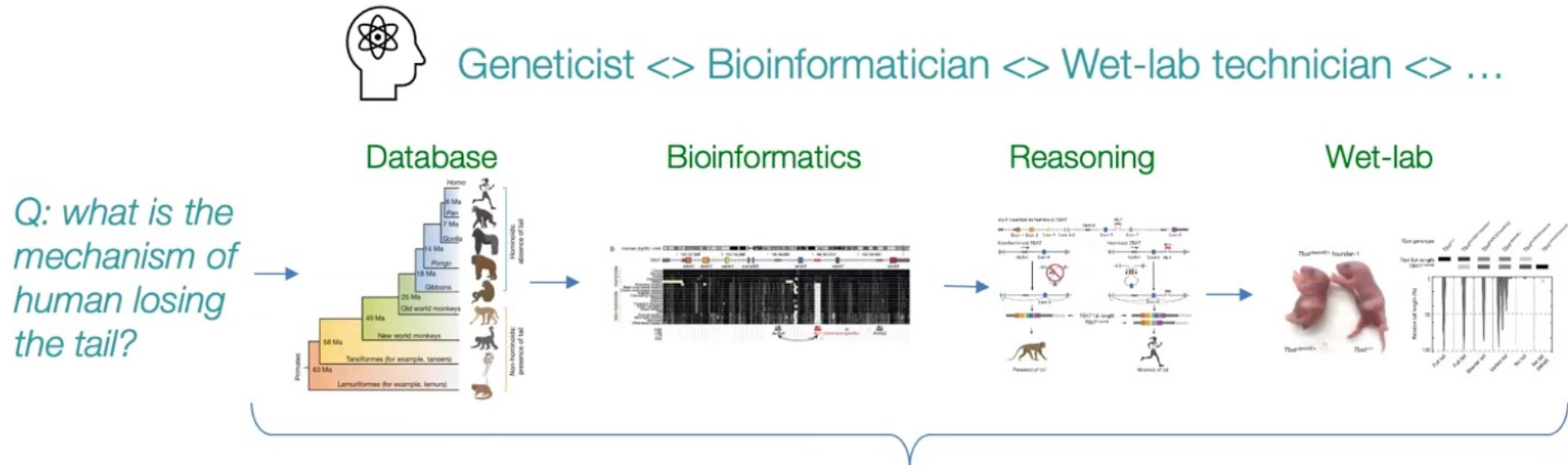
Knowledge is knowing a
tomato is a fruit.

Wisdom is not putting
it in a fruit salad.

— Miles Kington



Biomedical research workflows are complex, diverse, and abundant



It is intrinsically a loop of reasoning + actions!

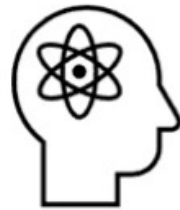
[4: Biomni talk: <https://www.youtube.com/watch?v=kOlrpy8V7GM> f]



How is biomedical research being done today?

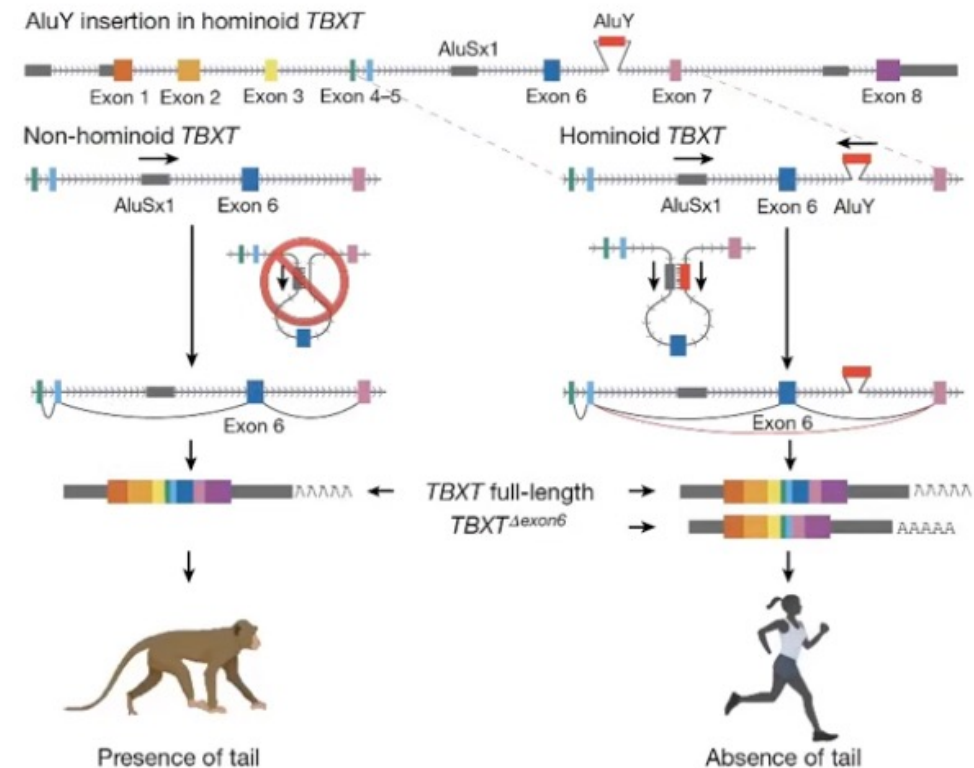


(1) Generating a hypothesis



biologist

“TBXT causes tail loss.”



How is biomedical research being done today?



(2) Validating a hypothesis



Biostatistician

Reasoning why a variant could cause a phenotype.

Statistical analysis from massive data generates p-values

GTExPortal

biobank^{uk}
Enabling scientific discoveries that improve human health

RCSB PDB
PROTEIN DATA BANK

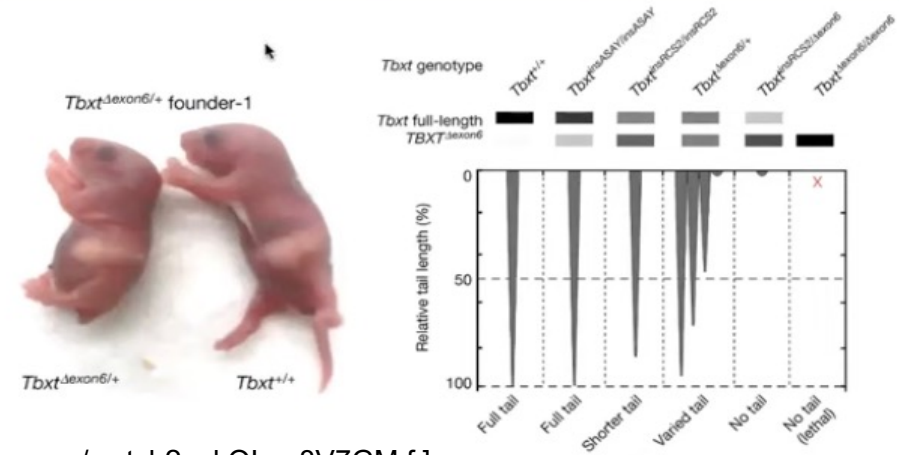
GWAS Catalog



biologist

A high throughput screen to search for genes that regulate a phenotype.

Targeted wet-lab experiment



Biomedical research at a glance



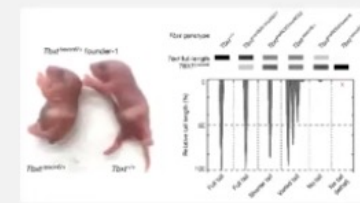
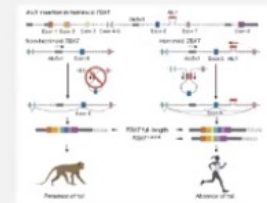
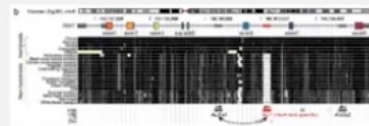
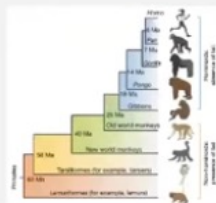
Outer-loop

Hypothesis generation

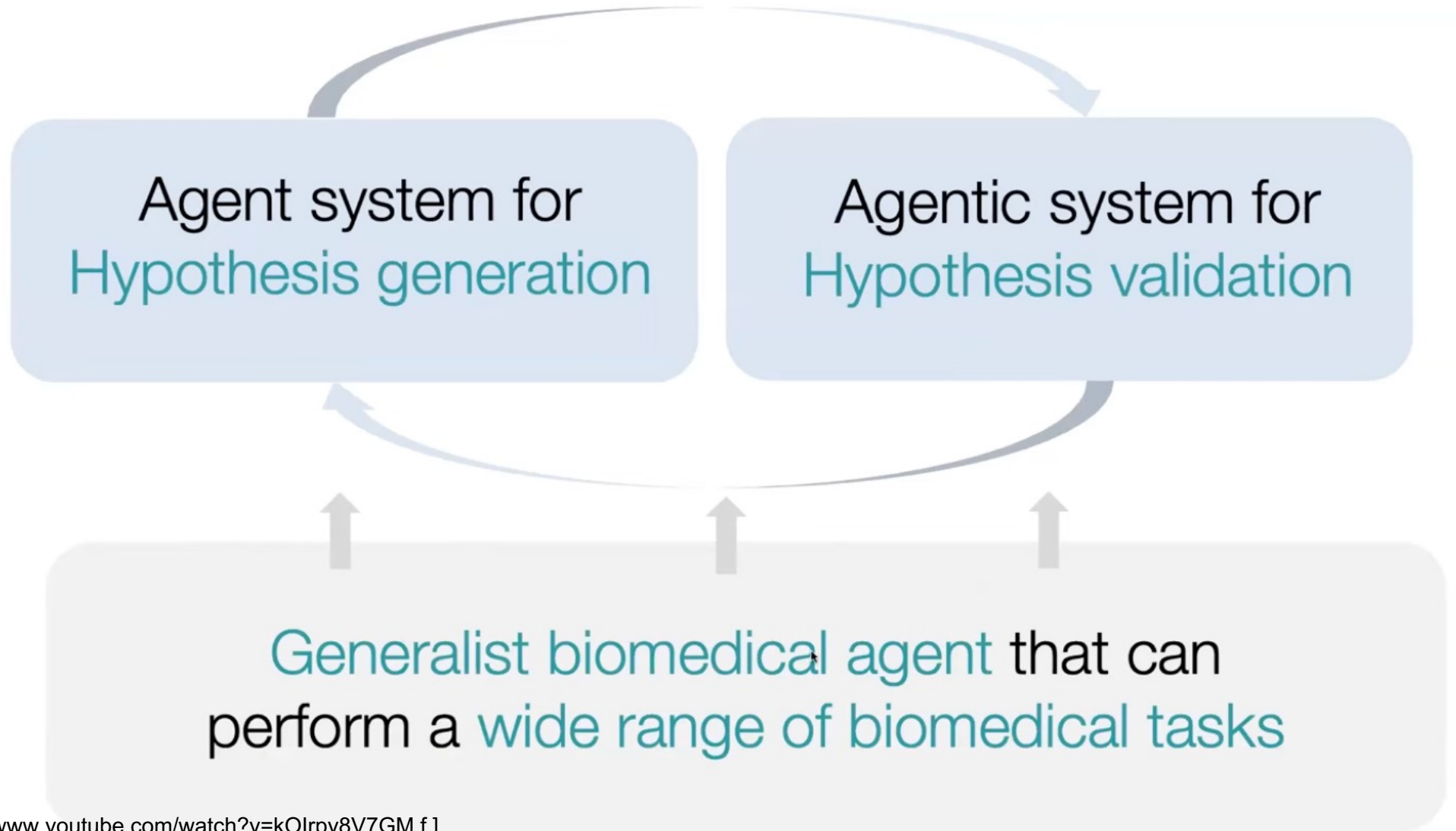
Hypothesis validation

Inner-loop

Massive number of individual workflows



A framework for building collaborative AI agents in biomedical research



In the quest for a biomedical research agent...

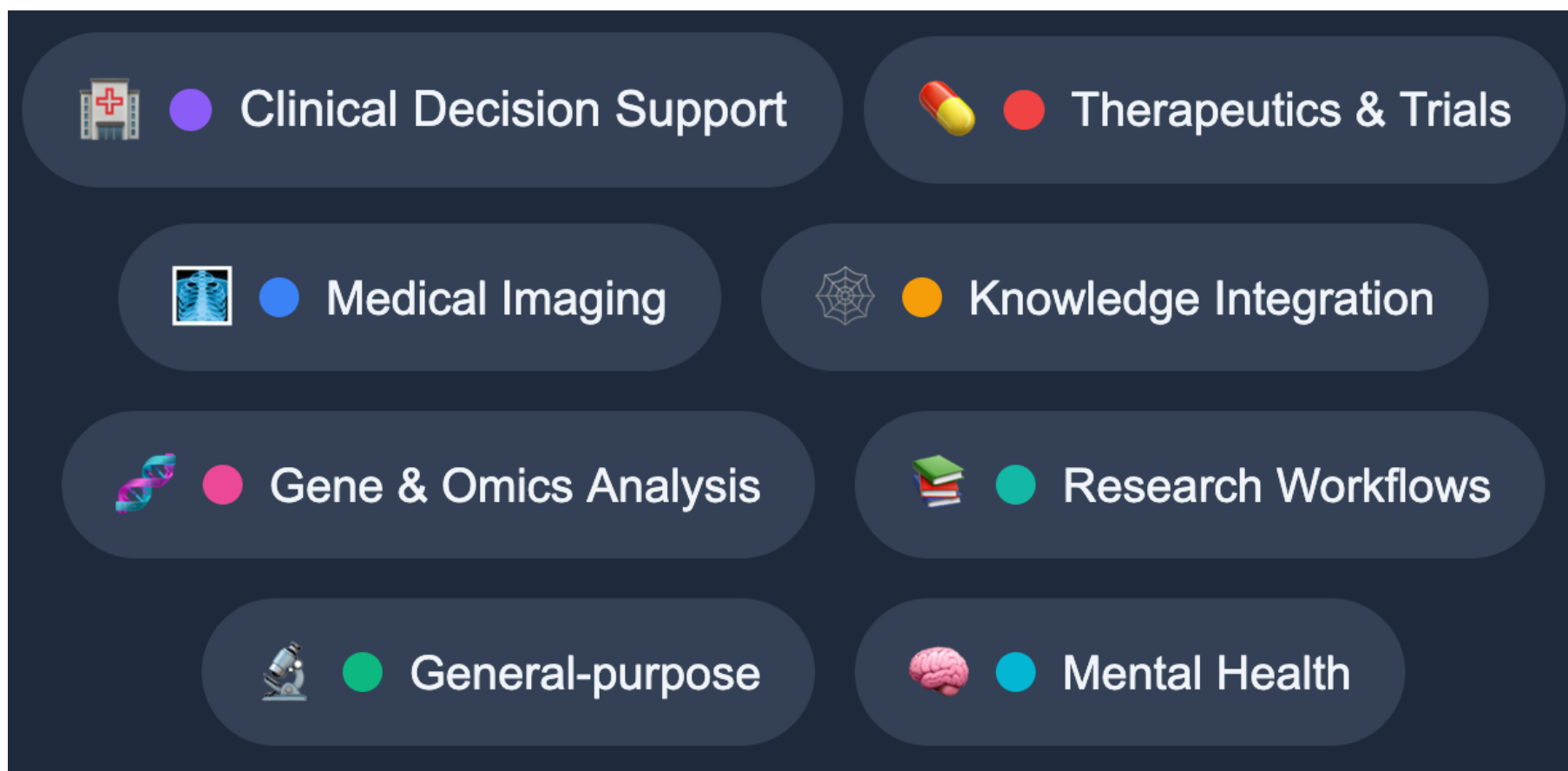


Capability	Details
Adapt	Plug in tools from genomics, drug discovery, clinical data, ... any domain
Unify	Orchestrate workflows across multiple domains
Evolve	Add new skills as research advances
Efficient	Automate repetitive data tasks
Reproducible	Transparent, executable record of every step
Scalable	Tackle complex, multi-step research at scale

The AI agent landscape in biomedical research (2024-2025)



Let's explore the key developments of the past two years...



[5]

Finding the right agent for the job



Task...	Consider...
Systematic literature reviews	AgentSLR
Analyze RNA-seq / scRNA-seq data	Medea, Biomni
Understand gene functions & pathways	GeneAgent
Analyze chest X-rays	MedRAX
Research drug interactions & polypharmacy	TxAgent
Query biomedical knowledge graphs	BioChatter, ESCARGOT
General biomedical research tasks	BioChatter, Biomni
Clinical decision support	MDAgents

Finding the right agent for the job



Task...	Consider...
Histopathology diagnosis	PathFinder
EHR reasoning & clinical tasks	MedAgentBench
Match patients to clinical trials	TrialMatchAI
End-to-end research automation	OpenLens AI
Mental health / psychotherapy	AI Psychotherapy
Evaluate clinical AI agents	AgentClinic

Getting started: no-code web-based tools



BioChatter



Biomni*



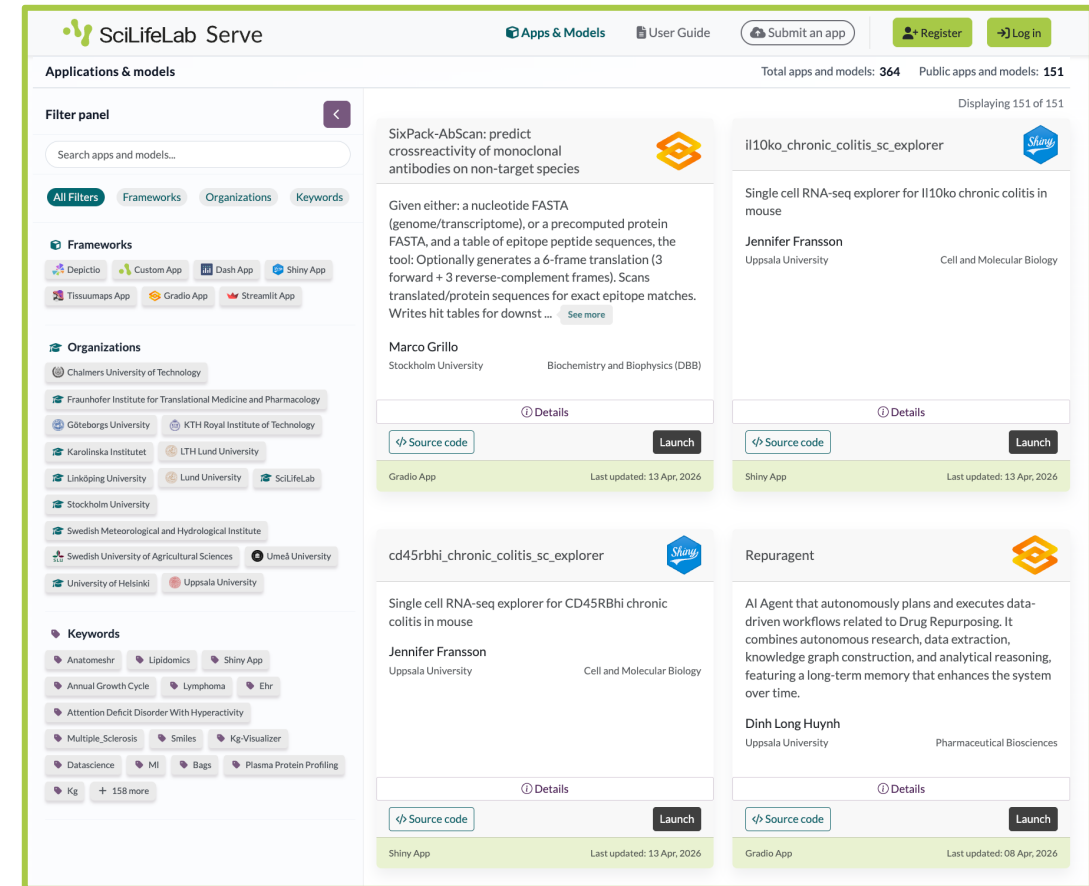
GeneAgent

*Biomni is available via the **SciLifeLab Serve** platform.
Contact: serve@scilifelab.se

SciLifeLab Serve: serve.scilifelab.se



- Hosting of apps and trained ML models for life science researchers.
- Graphical user interface or API endpoints.
- Available free of charge to all affiliated with a Swedish research institution.
- Currently ≈ 350 apps and models hosted in total.
- Use cases:
 - Allow reviewers/colleagues to verify the results/explore the behaviour of your model
 - Use for predictions with new input data for researchers, government agencies, general public.
 - Demo app as a first step in commercialisation effort.



Source: Screenshot of <https://serve.scilifelab.se>

Privacy, ethics & responsible use



Your data type	Recommended approach	Why it matters
Human patient or clinical data	Install locally or use institutional servers	Keeps sensitive information within your firewall and compliant with institutional policies
Published or synthetic datasets	Web tools are fine	Public data carries no privacy risk
Collaborative multi-site studies	Check with your data access committee first	Ensures all institutions agree on AI tool usage

Legal Aspects of AI Agents



- **EU AI Act:** Healthcare AI is high-risk; requires transparency, human oversight, and conformity assessments
- **Data protection:** GDPR applies; ensure consent and caution with data leaving the EU
- **Liability:** Responsibility for AI errors; document AI decisions for accountability
- **Sector compliance:** Clinical/diagnostic AI may require medical device regulation (MDR/IVDR)

SciLifeLab AI Strategy

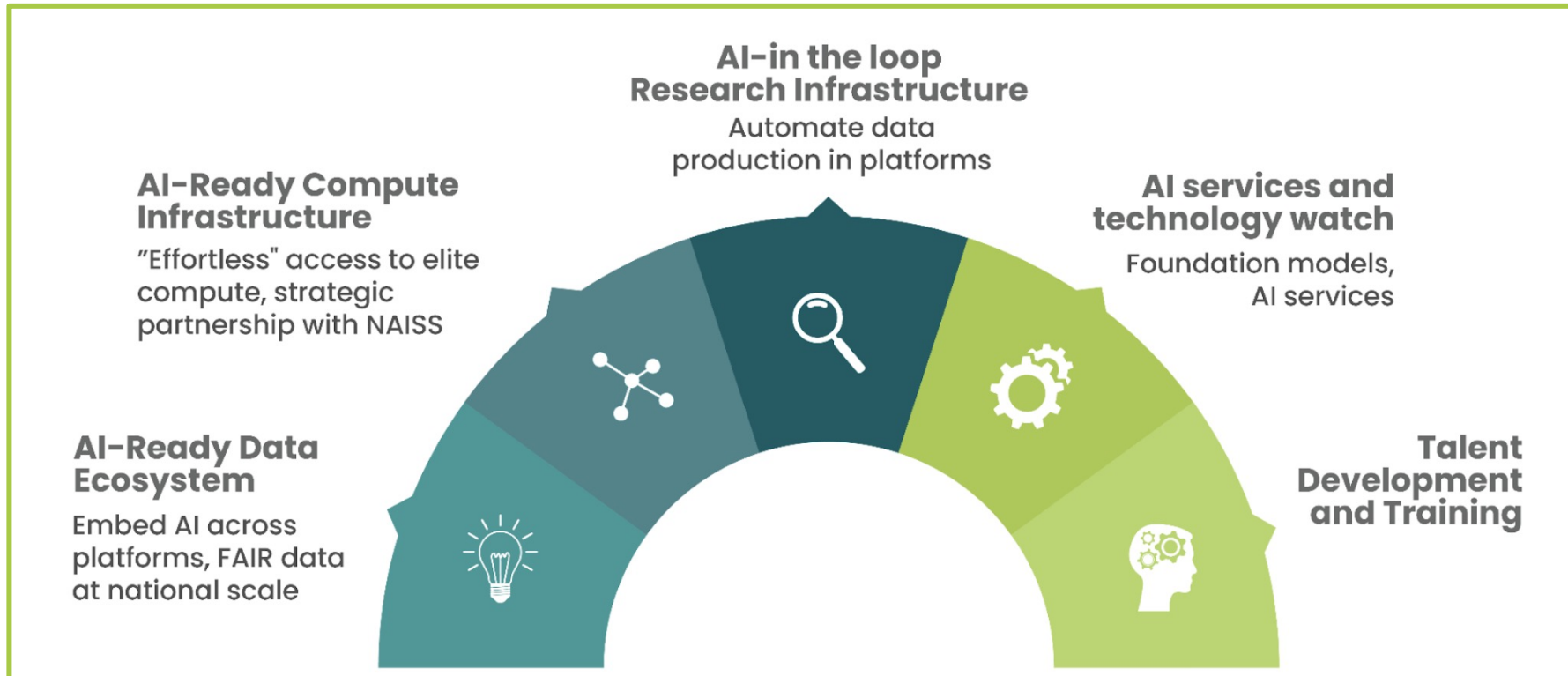


- SciLifeLab has a strong foundation for life science AI development and AI-driven discoveries.
- SciLifeLab AI Strategy identifies strategic pillars and initial priorities.
- v1.0 published on 2026.03.06
- To be translated into a roadmap and updated yearly.

<https://shorturl.at/7j8hU>



SciLifeLab AI Strategy Pillars



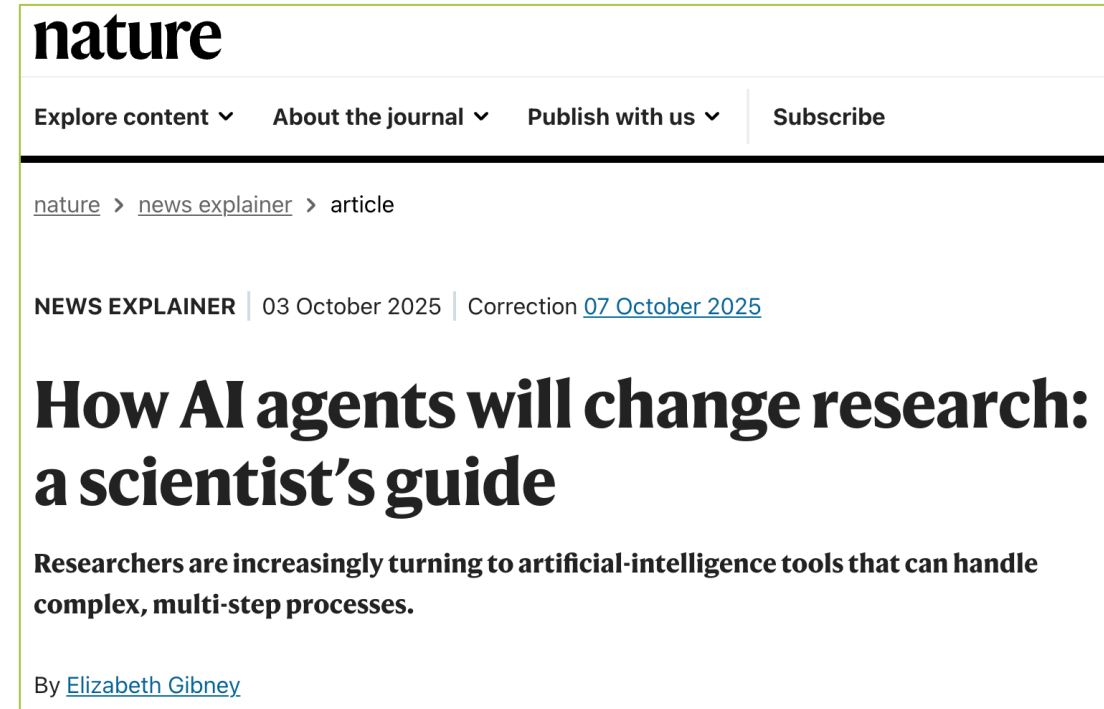
published on 2026.03.06

<https://shorturl.at/7j8hU>

Good practices for publication



- **Disclosure:** Journals appreciate transparency, mention “analysis pipeline designed with [Agent Name]” in methods
- **Reproducibility:** Save the agent's execution logs (most tools provide these) so others can follow your steps
- **Diversity awareness:** Many biomedical databases have ancestry biases, interpret results carefully if working with underrepresented populations



[6: Nature: <https://www.nature.com/articles/d41586-025-03246-7>]

Setting realistic expectations



- Agents are powerful partners, but not perfect (yet 😊)
- Common situations to navigate,

Situation	What to keep in mind
The agent suggests an unexpected analysis step	Ask <i>why</i> , good agents explain their reasoning; great ones show their sources
Results differ from your lab's previous findings	Agents query public databases that update regularly, check version dates
The output looks "too confident"	Your intuition is still the best quality control, Do not trust, always verify

Simple safeguards



- ✓ **Start with what you know:** Test the agent with a gene or pathway you understand well before exploring new territory
- ✓ **Check the timestamp:** Database queries reflect the last update cycle, verify if that fits your research timeline
- ✓ **Keep the human in the loop:** The agent handles execution; you handle interpretation

“ Think of it as a skilled research assistant; speeds up the workflow, but you still review the final draft before submission. ”

The five-minute comfort test



Before using any agent on your real research data:

1. Ask it something you already know (e.g., “What does BRCA1 do?”)
2. Check if the answer matches your knowledge and includes source links
3. If yes  → you're ready to try your actual research question

Emerging trends



Trend	Examples	Impact
Self-verification & hallucination reduction	GeneAgent, ESCARGOT, AgentSLR	Grounds scientific outputs in validated databases and literature
Multi-agent clinical collaboration	MDAgents, AI Psychotherapy, AgentClinic	Simulates specialist team consultations and clinical encounters
Tool-augmented therapeutic reasoning	TxAgent, Medea, Biomni	Grounds AI in validated databases
Multi-modal medical imaging	MedRAX, PathFinder	Orchestrates specialized vision models for radiology and pathology

Emerging trends



Trend	Examples	Impact
Knowledge graph integration	BioChatter, ESCARGOT	Structures disease-gene-drug relationships for grounded answers
EHR-native agent evaluation	MedAgentBench, AgentClinic	Benchmarks agents on realistic FHIR-based clinical tasks
Autonomous research workflows	OpenLens AI, AgentSLR, Medea	End-to-end pipelines from literature review to manuscript generation
Clinical trial matching	TrialMatchAI	Connects patients to relevant trials
Explainable mental health AI with ethical safeguards	AI Psychotherapy	Transparent step-by-step disorder predictions

What's next



- **Expanding tool ecosystems:** More databases, more analysis methods integrated
- **Wet-lab integration:** Agents that can design AND help execute experiments
- **Regulatory considerations:** As agents enter clinical settings, validation becomes crucial
- **Community building:** BioContextAI, open-source contributions
- **Multi-modal agents:** Combining text, images, molecular structures

What's next



- **Unprecedented speed:** Automate entire workflows, compressing weeks of work into hours.
- **Parallel exploration:** Tackle multiple complex questions and experimental pathways *simultaneously*.
- **Augmented creativity:** Break through cognitive barriers by exploring novel, data-driven hypotheses you might never have considered.

Concluding remarks: embracing the evolution



Where we stand today...

✗ AI agents are imperfect → they will frustrate you

✓ But progress is rapid: impossible → routine in six months

Concluding remarks: embracing the evolution



The researcher mindset...

- ✓ Stay skeptical, but be cautiously curious
- ✓ Early adopters can influence how these tools and their safeguards evolve.
- ✓ Better to guide the evolution than catch up later

Concluding remarks: embracing the evolution



Science has always adapted and evolved...

- ✓ Sanger sequencing → CRISPR
- ✓ Paper notebooks → electronic databases
- ✓ Adaptation is our core principle; change is our constant

Concluding remarks: embracing the evolution



Moving forward...

- ✅ Start small, test carefully, stay critical
- ✅ Don't step away from the conversation
- ✅ **The future belongs to those who adapt!** 😊

References: selected AI agents (alphabetical)



- i. AgentClinic: <https://github.com/samuelschmidgall/AgentClinic>
- ii. AgentSLR: <https://github.com/OxRML/AgentSLR>
- iii. AI Psychotherapy: https://github.com/mithatco/mental_health_multiagent
- iv. BioChatter: <https://github.com/biocypher/biochatter>
- v. Biomni: <https://github.com/snap-stanford/biomni>
- vi. ESCARGOT: <https://github.com/EpistasisLab/escargot>
- vii. GeneAgent: <https://github.com/ncbi-nlp/GeneAgent>
- viii. MDAgents: <https://github.com/mitmedialab/MDAgents>
- ix. Medea: <https://github.com/mims-harvard/Medea>
- x. MedAgentBench: <https://github.com/stanfordmlgroup/MedAgentBench>
- xi. MedRAX: <https://github.com/bowang-lab/MedRAX>
- xii. OpenLens AI: <https://github.com/jarrycyx/openlens-ai>
- xiii. PathFinder: <https://pathfinder-dx.github.io/>
- xiv. TrialMatchAI: <https://github.com/cbib/TrialMatchAI>
- xv. TxAgent: <https://github.com/mims-harvard/TxAgent>

References:



- [1] Cell: <https://www.sciencedirect.com/science/article/pii/S0092867424010705>
- [2] AI-generated by Mahbub using ChatGPT (5.2 Pro)
- [3] BioChatter slides: https://ftp.ebi.ac.uk/pub/training/2025/Webinars/20250310_LLM-driven_bioscience%e2%80%9494PDF.pdf
- [4] Biomni talk: <https://www.youtube.com/watch?v=kOIrpy8V7GM>
- [5] AI agent landscape tool by Mahbub: <https://shorturl.at/kxydE>
- [6] Nature: <https://www.nature.com/articles/d41586-025-03246-7>

Thank you! 😊



Contact: datacentre@scilifelab.se, serve@scilifelab.se

Explore the AI agent landscape tool 😊

Use the link below, or scan this QR code

<https://shorturl.at/kxydE>



Create your own landscape tool 😊

<https://shorturl.at/sRITg>



The future belongs to those who adapt! 😊